

Dynamics on the Berkovich Affine Line

Yifan Yao

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Abstract

In this paper, we give a self-contained introduction to dynamics on the Berkovich affine line over an algebraically closed complete non-Archimedean field. We first construct the Berkovich affine line, classify its points, and explain its tree structure and hyperbolic metric. We then study the extension of polynomials and rational maps, reduction, local degree, Fatou and Julia sets, potential theory, the equilibrium measure, and entropy. We explain how the resulting theorems of projective theory restrict back to affine theory. The central result is the non-Archimedean analogue of Brolin's theorem: outside the exceptional set, normalized preimages under a degree $d \geq 2$ polynomial equidistribute toward the equilibrium measure.

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1 Introduction

In classical complex dynamics we study the iterates

$$f^n = f \circ f \circ \cdots \circ f$$

of a polynomial or rational map. If $f : \mathbf{A}^1(\mathbb{C}) \rightarrow \mathbf{A}^1(\mathbb{C})$ is a polynomial of degree $d \geq 2$, then a central question is to understand the asymptotic behavior of the orbit

$$z, f(z), f^2(z), \dots$$

Fatou and Julia showed that the complex plane decomposes into two complementary regions. On the Fatou set the family of iterates is normal, so the dynamics is stable; on the Julia set the dynamics is chaotic. This theory is developed in detail in the books of Carleson–Gamelin [4] and Milnor [6].

Another fundamental result in complex polynomial dynamics is Brolin’s theorem [2]. If f is a polynomial of degree $d \geq 2$, let K_f be the filled Julia set, the set of points with bounded forward orbit, and let ρ_f be its harmonic measure. Then, for all points $a \in \mathbf{A}^1(\mathbb{C})$ except possibly one exceptional point,

$$d^{-n}(f^n)^*[a] \longrightarrow \rho_f$$

in the weak topology of probability measures. Equivalently, the preimages of a non-exceptional point become equidistributed with respect to a canonical invariant measure. Thus potential theory and dynamics are deeply linked.

The goal of this paper is to describe the analogue of this story over a non-Archimedean field. Throughout the paper, unless otherwise stated, K denotes an algebraically closed field, complete with respect to a non-trivial non-Archimedean absolute value $|\cdot|$, and of characteristic zero. A standard example is $K = \mathbb{C}_p$, the completion of an algebraic closure of $\mathbb{Q}_p$.

There is an immediate obstacle. The topological space K is totally disconnected, and it is usually not locally compact. Consequently, the ordinary affine line $\mathbf{A}^1(K)$ is too small for a good global theory of normal families, connected Julia sets, and limiting probability measures. The Berkovich affine line $\mathbf{A}_{\text{Berk}}^1$ solves this problem by adding many non-classical points. The resulting space contains K densely, is locally compact and Hausdorff, and is uniquely path connected. In other words, $\mathbf{A}_{\text{Berk}}^1$ is a genuine topological tree.

The references suggested for this project, especially Favre–Rivera-Letelier [5] and Baker–Rumely [3], work primarily on the Berkovich projective line $\mathbf{P}_{\text{Berk}}^1$. The projective line is obtained from the affine line by adding the point at infinity,

$$\mathbf{P}_{\text{Berk}}^1 = \mathbf{A}_{\text{Berk}}^1 \cup \{\infty\},$$

which is a compact space. Compactness is essential for probability measures, potential theory, entropy, and the formulation of rational dynamics. For a polynomial $f \in K[z]$, the point ∞ is a superattracting fixed point, and the interesting affine dynamics is contained in the complement of its basin of attraction. Therefore the projective theory is the correct global framework, while the affine theory is obtained by restricting it to $\mathbf{A}_{\text{Berk}}^1$.

This paper is organized as follows. In section 2 we will construct the Berkovich affine line, classify its four types of points, and explain its tree structure. In section 3 we will discuss the

passage from the affine line to the projective line and explain why the projective compactification is unavoidable in the references. In section 4 we will develop elementary dynamics: extension of maps, local degree, images of disks, and good reduction. In section 5 we will introduce Fatou and Julia sets. In sections 6 and 7 we will give the potential-theoretic construction of the equilibrium measure and state the equidistribution theorem. In section 8 and 9, we will summarize the entropy theory and give examples respectfully from [5].

2 The Berkovich affine line

Although in class we have seen lots of results mentioned in this section, for self-containedness, let us recall the beginning of Berkovich's story.

2.1 Non-Archimedean preliminaries

Let K be as above. The absolute value is called non-Archimedean if it satisfies the strong triangle inequality

$$|x + y| \leq \max\{|x|, |y|\}$$

for all $x, y \in K$. This simple inequality has many consequences. A frequently used one is: if $|x| \neq |y|$, then

$$|x + y| = \max\{|x|, |y|\}.$$

Closed disks are defined by

$$D(a, r) = \{z \in K : |z - a| \leq r\}, \quad a \in K, r \geq 0.$$

Open disks are defined similarly by $D(a, r)^- = \{z : |z - a| < r\}$. Because of the ultrametric inequality, every point of a disk is a center: if $b \in D(a, r)$, then $D(a, r) = D(b, r)$. Moreover, two disks are either disjoint or one contains the other.

The set of possible radii is $|K^\times| \subset \mathbb{R}_{>0}$. Since K is algebraically closed, this value group is divisible, but it need not be all of $\mathbb{R}_{>0}$. This distinction is one reason why Berkovich points of type III appear below.

Definition 2.1. For a polynomial $P(T) = \sum_i c_i(T - a)^i \in K[T]$ and a number $r \geq 0$, define

$$|P|_{a,r} = \max_i |c_i| r^i.$$

This is the Gauss norm of P on the disk $D(a, r)$.

If $r \in |K^\times|$, then $|P|_{a,r}$ is the supremum of $|P(z)|$ over $z \in D(a, r)$. Even when $r \notin |K^\times|$, the formula still defines a multiplicative seminorm on $K[T]$ extending the absolute value on K .

2.2 Definition of $\mathbf{A}_{\text{Berk}}^1$

Definition 2.2. The Berkovich affine line over K is the set

$$\mathbf{A}_{\text{Berk}}^1 = \{\text{multiplicative seminorms } [\cdot]_x : K[T] \rightarrow \mathbb{R}_{\geq 0} \text{ extending } |\cdot| \text{ on } K\}.$$

Thus each point $x \in \mathbf{A}_{\text{Berk}}^1$ is a function $P \mapsto [P]_x$ such that

- (i) $[0]_x = 0$ and $[1]_x = 1$;
- (ii) $[PQ]_x = [P]_x[Q]_x$;
- (iii) $[P + Q]_x \leq \max\{[P]_x, [Q]_x\}$;
- (iv) $[c]_x = |c|$ for all $c \in K$.

The topology on $\mathbf{A}_{\text{Berk}}^1$ is the weakest topology for which every evaluation map

$$x \mapsto [P]_x, \quad P \in K[T],$$

is continuous. This is called Gel'fand topology.

Every classical point $a \in K$ gives a Berkovich point, still denoted a , by

$$[P]_a = |P(a)|.$$

This embeds K into $\mathbf{A}_{\text{Berk}}^1$. The image is dense, but it is only a small part of the whole Berkovich line. For every disk $D(a, r)$ the Gauss norm $|\cdot|_{a,r}$ gives a point, which we denote by $\zeta_{a,r}$. If $r = 0$, this is the classical point a .

Proposition 2.3. *The space $\mathbf{A}_{\text{Berk}}^1$ is Hausdorff, locally compact, and path connected. The classical affine line K is dense in $\mathbf{A}_{\text{Berk}}^1$.*

Proof. We give the main ideas. The topology is defined by the family of real-valued functions $x \mapsto [P]_x$. Distinct seminorms differ on some polynomial, and hence may be separated by open intervals in $\mathbb{R}_{\geq 0}$. This proves the Hausdorff property. Local compactness is proved by identifying compact subsets with inverse limits of compact sets of seminorms bounded by prescribed radii, which is one of the basic compactness results in Berkovich's construction. Path connectedness follows more concretely from the disk description below: two disk points are joined by increasing or decreasing the radius until the corresponding disks meet, and arbitrary points are limits of such disk points. Density of K follows from the fact that every disk seminorm is the limit, in the Gel'fand topology, of classical points in sufficiently large finite extensions. Since K is algebraically closed, these points already lie in K . $\square$

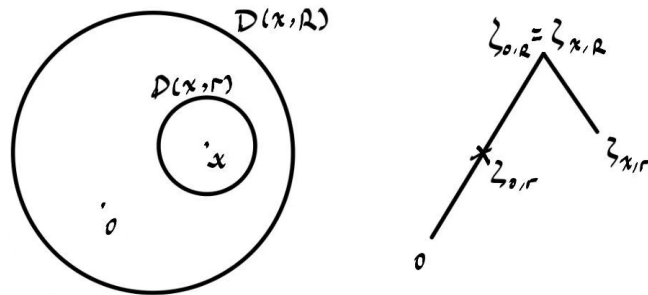


Figure 1: A path from 0 to $\zeta_{x,r}$

2.3 Classification of points

The classification of points is one of the most useful features of $\mathbf{A}_{\text{Berk}}^1$ and is emphasized in class. It states that every point is represented by either a disk or a nested family of disks.

Theorem 2.4 (Classification of Berkovich affine points). *Every point of $\mathbf{A}_{\text{Berk}}^1$ is of exactly one of the following four types.*

- Type I. *A classical point $a \in K$, corresponding to the seminorm $P \mapsto |P(a)|$. Equivalently this is the disk point $\zeta_{a,0}$.*
- Type II. *A disk point $\zeta_{a,r}$ where $r \in |K^\times|$. These points correspond to closed disks whose radius lies in the value group.*
- Type III. *A disk point $\zeta_{a,r}$ where $r \notin |K^\times|$. These points correspond to closed disks of irrational radius with respect to the value group.*
- Type IV. *A point determined by a decreasing sequence of closed disks*

$$D(a_1, r_1) \supset D(a_2, r_2) \supset \cdots$$

with empty intersection in K . The associated seminorm is

$$[P]_x = \lim_{n \rightarrow \infty} |P|_{a_n, r_n}.$$

Such points occur only when K is not spherically complete.

Proof. Given a multiplicative seminorm x on $K[T]$, define its diameter by

$$\text{diam}(x) = \inf_{a \in K} [T - a]_x.$$

For each $a \in K$ and $r \geq 0$, the disk seminorm $\zeta_{a,r}$ has diameter r . One associates to x the family of disks

$$\mathcal{D}_x = \{D(a, r) : [T - a]_x \leq r\}.$$

The ultrametric inequality implies that this family is totally ordered by inclusion. If the intersection consists of one point a , then x is evaluation at a , giving type I. If the intersection is a closed disk $D(a, r)$ with $r > 0$, then $x = \zeta_{a,r}$; according as r lies in $|K^\times|$ or not, the point has type II or type III. Finally, if the corresponding decreasing family of disks has empty intersection, it defines a limiting seminorm of type IV. The four alternatives are mutually exclusive by the value of the diameter and by whether the associated nested disks have a point of intersection. $\square$

Remark 2.5. Type II points are the branching points of the Berkovich tree. Type III points lie in the interior of edges. Type IV points are endpoints that are not classical. The type II point $\zeta_{0,1}$ is called the Gauss point and will be denoted ζ_G .

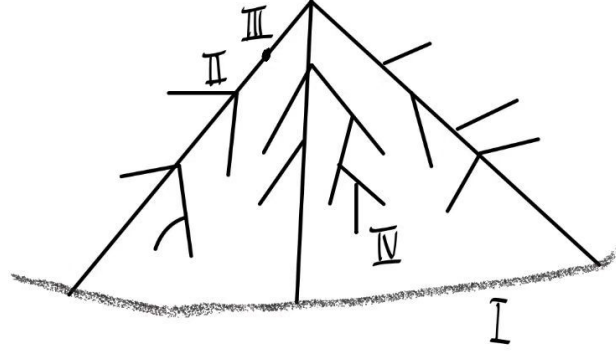


Figure 2: A rough picture of Berkovich affine line

2.4 The partial order and tree structure

There is a natural partial order on $\mathbf{A}_{\text{Berk}}^1$.

Definition 2.6. For $x, y \in \mathbf{A}_{\text{Berk}}^1$, write $x \preceq y$ if

$$[P]_x \leq [P]_y \quad \text{for every } P \in K[T].$$

For disk points this means exactly that the corresponding disks are nested:

$$\zeta_{a,r} \preceq \zeta_{b,s} \iff D(a,r) \subseteq D(b,s).$$

Given two points $x, y \in \mathbf{A}_{\text{Berk}}^1$, there is a smallest point $x \vee y$ that dominates both. For classical points $a, b \in K$, this point is

$$a \vee b = \zeta_{a,|a-b|} = \zeta_{b,|a-b|},$$

the smallest closed disk containing both a and b .

Theorem 2.7 (Tree property). *The Berkovich affine line $\mathbf{A}_{\text{Berk}}^1$ is uniquely path connected. For any two points x, y , there is a unique embedded path $[x, y]$ joining them. If $x \vee y$ denotes their least common upper bound, then*

$$[x, y] = [x, x \vee y] \cup [x \vee y, y],$$

where each segment is linearly ordered by the partial order.

Proof. For disk points the proof is explicit. If $D(a, r) \subseteq D(b, s)$, then the path from $\zeta_{a,r}$ to $\zeta_{b,s}$ is

$$t \mapsto \zeta_{a,t}, \quad r \leq t \leq s,$$

with the understanding that the center may be changed without changing the disk. If neither disk contains the other, first increase the radius of $D(a, r)$ until reaching the smallest disk containing both disks, and then decrease along the other branch. This gives a path through $\zeta_{a,r} \vee \zeta_{b,s}$. The

fact that any two disks are either disjoint or nested ensures uniqueness. General Berkovich points are limits of disk points or endpoints of nested disk chains, and the same ordered description gives the unique path. If a second path existed, its union with the first would contain a simple loop, contradicting the total ordering of disks along every branch. $\square$

The space therefore resembles a real tree. At a type II point $\zeta_{a,r}$, the connected components of $\mathbf{A}_{\text{Berk}}^1 \setminus \{\zeta_{a,r}\}$ are called directions at $\zeta_{a,r}$. If $r \in |K^\times|$, choose $b \in K$ with $|b| = r$. After the change of coordinate $z \mapsto (z - a)/b$, the directions at $\zeta_{a,r}$ are naturally indexed by $\mathbb{P}^1(k)$, where k is the residue field. The direction corresponding to ∞ points away from the disk $D(a, r)$, and the other directions correspond to residue classes inside the disk.

2.5 The hyperbolic metric

The classical points are endpoints of the Berkovich tree. The hyperbolic space is obtained by removing them.

Definition 2.8. The affine Berkovich hyperbolic space is

$$\mathbb{H}_{\mathbf{A}} = \mathbf{A}_{\text{Berk}}^1 \setminus K.$$

For a disk point $\zeta_{a,r}$ with $r > 0$, put $\text{diam}(\zeta_{a,r}) = r$. If $x \preceq y$ are non-classical points, define

$$d_{\mathbb{H}}(x, y) = \log \frac{\text{diam}(y)}{\text{diam}(x)}.$$

For arbitrary non-classical points x, y , define

$$d_{\mathbb{H}}(x, y) = d_{\mathbb{H}}(x, x \vee y) + d_{\mathbb{H}}(y, x \vee y).$$

This metric is finite on $\mathbb{H}_{\mathbf{A}}$ for type II and III points and extends naturally to suitable type IV points when their diameters are positive. In practice, most potential-theoretic arguments are made on finite subtrees spanned by type II or type III points and then extended by limiting arguments.

Example 2.9. For $0 < r \leq s$, the distance between $\zeta_{0,r}$ and $\zeta_{0,s}$ is $\log(s/r)$. Thus the segment from 0 to the Gauss point $\zeta_G = \zeta_{0,1}$ has infinite length if one includes the classical endpoint 0, since $\log(1/r) \rightarrow \infty$ as $r \rightarrow 0$.

The hyperbolic metric is not the same as the Gel'fand topology. The Gel'fand topology is compact-like and weak, while the hyperbolic metric measures length along the tree. Both are needed: compactness and weak convergence use the Gel'fand topology, while Laplacians and potentials are naturally described by slopes along the hyperbolic tree.

Remark 2.10. Here is an interesting story about cutting K out of Berkovich affine or projective line. In physics we usually fix an upper energy scale Λ called UV cutoff. By doing this, we can kill the divergences lurking in the process of computing Feynman diagrams. In p-adic AdS/CFT duality, the object we deal with is so-called Bruhat-Tits tree, and there we also need to fix a cutoff Λ to avoid divergences. Furthermore, since the potential theory has been nicely built on the Berkovich space, there is no doubt that we can consider an AdS/CFT theory on Berkovich space likewise. I haven't seen others done this work before, or I might just miss it out.

3 From the affine line to the projective line

3.1 Why the projective line appears

The most powerful references use $\mathbf{P}_{\text{Berk}}^1$ instead of $\mathbf{A}_{\text{Berk}}^1$. The reason is: potential theory and invariant probability measures behave best on compact spaces, and $\mathbf{A}_{\text{Berk}}^1$ is not compact. The Berkovich projective line is compact, and rational maps naturally act on it.

Definition 3.1. The Berkovich projective line is the analytification of $\mathbf{P}_K^1$. It may be described as

$$\mathbf{P}_{\text{Berk}}^1 = \mathbf{A}_{\text{Berk}}^1 \cup \{\infty\},$$

where neighborhoods of ∞ are complements of sufficiently large closed Berkovich disks. Equivalently, it is obtained by gluing two copies of the Berkovich affine line by the coordinate change $w = 1/z$ on $K^\times$.

The point ∞ is a classical type I point. The projective line is compact, Hausdorff, and uniquely path connected. The hyperbolic space of the projective line is

$$\mathbb{H} = \mathbf{P}_{\text{Berk}}^1 \setminus \mathbf{P}^1(K).$$

Many formulas in potential theory are symmetric only on $\mathbf{P}_{\text{Berk}}^1$, because there is no preferred affine coordinate.

Proposition 3.2. Every polynomial $f \in K[z]$ of degree $d \geq 1$ extends uniquely to a continuous map

$$f : \mathbf{A}_{\text{Berk}}^1 \longrightarrow \mathbf{A}_{\text{Berk}}^1.$$

It further extends to a continuous map

$$f : \mathbf{P}_{\text{Berk}}^1 \longrightarrow \mathbf{P}_{\text{Berk}}^1$$

with $f(\infty) = \infty$. If $d \geq 2$, then ∞ is a superattracting fixed point.

Proof. For $x \in \mathbf{A}_{\text{Berk}}^1$ define $f(x)$ by

$$[P]_{f(x)} = [P \circ f]_x, \quad P \in K[T].$$

This is again a multiplicative seminorm extending $|\cdot|$, so it defines a point of $\mathbf{A}_{\text{Berk}}^1$. Continuity follows directly from the definition of the Gel'fand topology: the evaluation of P at $f(x)$ is the evaluation of $P \circ f$ at x . In the coordinate $w = 1/z$ near infinity, a degree d polynomial has the form

$$w \longmapsto \frac{1}{f(1/w)} = cw^d + O(w^{d+1})$$

with $c \neq 0$. Hence $w = 0$, which is $z = \infty$, is fixed with local degree d . □

3.2 Affine dynamics as part of projective dynamics

For a polynomial f , the affine and projective viewpoints are related as follows. Define the basin of infinity by

$$\mathcal{B}_\infty(f) = \{x \in \mathbf{P}_{\text{Berk}}^1 : f^n(x) \rightarrow \infty\}.$$

The filled Julia set is

$$K_f = \mathbf{P}_{\text{Berk}}^1 \setminus \mathcal{B}_\infty(f).$$

Since ∞ is fixed and attracting, $\mathcal{B}_\infty(f)$ is an open subset of the Fatou set. The set K_f is compact and contained in $\mathbf{A}_{\text{Berk}}^1$.

In the complex case, harmonic measure of K_f is constructed using the Green's function on the plane. In the non-Archimedean case, the corresponding measure is most naturally constructed on $\mathbf{P}_{\text{Berk}}^1$ and then observed to be supported in $\mathbf{A}_{\text{Berk}}^1$ for polynomials.

Proposition 3.3. *Let $f \in K[z]$ have degree $d \geq 2$, and let μ_f be its equilibrium measure (we will define it later) on $\mathbf{P}_{\text{Berk}}^1$. Then*

$$\mu_f(\mathcal{B}_\infty(f)) = 0,$$

and therefore $\text{supp}(\mu_f) \subseteq K_f \subset \mathbf{A}_{\text{Berk}}^1$.

Proof. The support of μ_f is the Berkovich Julia set, as proved later in Theorem 7.2. Since ∞ lies in an attracting Fatou component, its basin $\mathcal{B}_\infty(f)$ is contained in the Fatou set. Hence it is disjoint from the Julia set and has μ_f -measure zero. Alternatively, one may use the invariance $f_*\mu_f = \mu_f$ and the fact that points in the basin eventually enter arbitrarily small neighborhoods of ∞ , while μ_f has no atom at the attracting classical point ∞ . $\square$

Thus, although the construction is projective, the measure controlling polynomial dynamics is an affine object.

3.3 The affine-projective dictionary

We record explicitly how an affine statement should be translated into the projective language used in the references. Let $f \in K[z]$ be a polynomial of degree $d \geq 2$. As a rational map on $\mathbf{P}_{\text{Berk}}^1$, f has a totally invariant point at infinity:

$$f^{-1}(\infty) = \{\infty\}.$$

Thus ∞ is always an exceptional point. The remaining affine dynamics is obtained by ignoring the attracting basin of this point. More precisely, the following table summarizes the correspondence:

Affine polynomial language	Projective Berkovich language
$\mathbf{A}_{\text{Berk}}^1$	$\mathbf{P}_{\text{Berk}}^1 \setminus \{\infty\}$
points with bounded orbit	$\mathbf{P}_{\text{Berk}}^1 \setminus \mathcal{B}_\infty(f)$
affine Julia set	$\mathcal{J}_f \cap \mathbf{A}_{\text{Berk}}^1$
Brolin measure	μ_f on $\mathbf{P}_{\text{Berk}}^1$, supported in $\mathbf{A}_{\text{Berk}}^1$
exceptional affine point	$E_f \setminus \{\infty\}$.

There is a small but important subtlety in this table. The equality

$$\mathbf{P}_{\text{Berk}}^1 = \mathbf{A}_{\text{Berk}}^1 \cup \{\infty\}$$

should not be interpreted as saying that the local geometry near infinity is trivial. Rather, a neighborhood of ∞ contains many non-classical points: for example, the points corresponding in the coordinate $w = 1/z$ to small disks around $w = 0$. These are not new points beyond $\mathbf{A}_{\text{Berk}}^1$ plus ∞ ; they are the complements of large affine disks and are already visible from the affine coordinate as points far out on the branch toward infinity. Thus the projective line compactifies the affine tree by adding one classical endpoint and by making the branch toward infinity compact in the Gel'fand topology.

This viewpoint also explains why potential functions usually have a pole or normalization at infinity in the polynomial case. On the affine line one wants to measure escape to infinity; on the projective line this becomes a potential whose Laplacian is a difference of two probability measures, typically

$$\mu_f - [\infty].$$

The total mass is zero, as required for a Laplacian on a compact space. If one tried to work only on $\mathbf{A}_{\text{Berk}}^1$, the term $[\infty]$ would disappear from the space, and the Laplacian equation would lose its natural normalization. This is the technical reason that Baker–Rumely's potential theory is formulated on $\mathbf{P}_{\text{Berk}}^1$.

Finally, the projective formulation is coordinate-free. A polynomial becomes a polynomial only after choosing the affine chart $\mathbf{P}^1 \setminus \{\infty\}$. A conjugate $M \circ f \circ M^{-1}$ by an element of $\text{PGL}_2(K)$ may no longer be a polynomial in the original coordinate, but it defines the same dynamical system in a different projective coordinate. Good reduction, potential good reduction, local degree, the equilibrium measure, and entropy are all naturally invariant under such projective conjugacy. The affine theory is therefore a distinguished chart of a projective theory, not a separate theory.

4 Elementary dynamics on Berkovich spaces

4.1 Extension of rational maps

Let $\phi \in K(z)$ be a nonconstant rational function. Write $\phi = F/G$, where $F, G \in K[z]$ are relatively prime. The map ϕ is not everywhere defined on $\mathbf{A}^1(K)$, because poles are sent to infinity. On $\mathbf{P}^1(K)$, however, it is a genuine self-map. The Berkovich extension has the same property.

Theorem 4.1. *Every nonconstant rational function $\phi \in K(z)$ extends uniquely to a continuous, open, surjective map*

$$\phi : \mathbf{P}_{\text{Berk}}^1 \longrightarrow \mathbf{P}_{\text{Berk}}^1.$$

If ϕ has degree d , then every point has exactly d preimages counted with local degree.

Proof. On an affine chart where ϕ has no pole, one defines $\phi(x)$ by pullback of seminorms, as for polynomials. Near a pole one uses the coordinate $w = 1/z$ on the target. These definitions

agree on overlaps because they are induced by the same homomorphism of coordinate rings. The resulting map is continuous by the definition of the Gel'fand topology. Openness, surjectivity, and the degree formula follow from the corresponding finite morphism of analytic curves. Locally, after choosing coordinates, the map is represented by a finite analytic algebra, and the sum of the multiplicities in a fiber is the degree of the induced extension of function fields. $\square$

4.2 Local degree

Definition 4.2. Let $\phi \in K(z)$ be nonconstant and let $x \in \mathbf{P}_{\text{Berk}}^1$. The local degree of ϕ at x , denoted $\deg_x(\phi)$, is the multiplicity with which ϕ maps a sufficiently small Berkovich neighborhood of x onto a neighborhood of $\phi(x)$. At a classical point $a \in \mathbf{P}^1(K)$ this is the usual ramification index.

For a type II point x , local degree can be computed by reduction. After changing coordinates so that x and $\phi(x)$ are both the Gauss point, the map has a normalized representation $\phi = F/G$ with coefficients in the valuation ring and at least one coefficient of absolute value 1. Reducing coefficients modulo the maximal ideal gives a rational map $\tilde{\phi}$ over the residue field. Then

$$\deg_x(\phi) = \deg(\tilde{\phi}).$$

Proposition 4.3 (Degree formula). *For every $y \in \mathbf{P}_{\text{Berk}}^1$,*

$$\sum_{x \in \phi^{-1}(y)} \deg_x(\phi) = \deg(\phi).$$

Moreover, local degrees are multiplicative under composition:

$$\deg_x(\phi \circ \psi) = \deg_x(\psi) \deg_{\psi(x)}(\phi).$$

Proof. The first formula is the analytic form of the algebraic fact that a finite morphism of degree d has fibers of total multiplicity d . It may also be verified on sufficiently small affinoid neighborhoods, where the map is described by a finite algebra of rank d . The second formula follows immediately from composing the corresponding finite local extensions. $\square$

4.3 Images of disks

Polynomials over non-Archimedean fields behave particularly well on disks.

Proposition 4.4. *Let $f \in K[z]$ and let $D(a, r)$ be a closed disk. Then $f(D(a, r))$ is a closed disk. More precisely, if*

$$f(z) = \sum_{i=0}^m c_i (z - a)^i,$$

then

$$f(D(a, r)) = D(f(a), R), \quad R = \max_{i \geq 1} |c_i| r^i.$$

Consequently,

$$f(\zeta_{a,r}) = \zeta_{f(a),R}.$$

Proof. For $z \in D(a, r)$, the ultrametric inequality gives

$$|f(z) - f(a)| \leq \max_{i \geq 1} |c_i| r^i = R.$$

Thus the image is contained in $D(f(a), R)$. Conversely, the non-Archimedean maximum modulus principle and the Weierstrass preparation theorem imply that every value in this disk is achieved, counted with multiplicity, by a point of $D(a, r)$. For disk seminorms this is exactly the identity $[P]_{f(\zeta_{a,r})} = [P \circ f]_{a,r}$. $\square$

Example 4.5. For $f(z) = z^d$, one has

$$f(\zeta_{a,r}) = \zeta_{a^d, R}, \quad R = \max_{1 \leq i \leq d} \binom{d}{i} |a|^{d-i} r^i.$$

In particular, on the branch from 0 to ∞ ,

$$f(\zeta_{0,r}) = \zeta_{0,r^d}.$$

Thus the hyperbolic distance from the Gauss point is multiplied by d along this branch.

4.4 Reduction and good reduction

Let $\mathcal{O}_K = \{x \in K : |x| \leq 1\}$ be the valuation ring, $\mathfrak{m}_K = \{x : |x| < 1\}$ its maximal ideal, and $\tilde{K} = \mathcal{O}_K / \mathfrak{m}_K$ the residue field.

Let $\phi \in K(z)$ have degree d . Choose a homogeneous representation

$$\phi([X : Y]) = [F(X, Y) : G(X, Y)]$$

where F and G are homogeneous degree d forms with no common factor. After multiplying both by a scalar, we may assume that all coefficients lie in $\mathcal{O}_K$ and at least one coefficient has absolute value 1. Reducing coefficients modulo $\mathfrak{m}_K$ gives homogeneous forms $\tilde{F}, \tilde{G}$ over $\tilde{K}$.

Definition 4.6. The map ϕ has good reduction if $\tilde{F}$ and $\tilde{G}$ have no common factor and the reduced map

$$\tilde{\phi} = [\tilde{F} : \tilde{G}]$$

has degree d . It has potentially good reduction if some conjugate $M \circ \phi \circ M^{-1}$ by $M \in \mathrm{PGL}_2(K)$ has good reduction.

Proposition 4.7. A rational map ϕ has good reduction if and only if the Gauss point ζ_G is totally invariant with full local degree:

$$\phi^{-1}(\zeta_G) = \{\zeta_G\}, \quad \deg_{\zeta_G}(\phi) = \deg(\phi).$$

It has potentially good reduction if and only if there exists a type II point x such that

$$\phi^{-1}(x) = \{x\}.$$

Proof. For the Gauss point, the action of ϕ on the tangent space is exactly the reduction $\tilde{\phi}$ on $\mathbb{P}^1(\tilde{K})$. Good reduction means that no degree is lost after reduction, i.e. $\deg(\tilde{\phi}) = \deg(\phi)$. This is equivalent to the Gauss point mapping to itself with full local degree and no other preimage. Conjugating by an element of $\mathrm{PGL}_2(K)$ sends the Gauss point to an arbitrary type II point, giving the potentially good reduction criterion. $\square$

Good reduction is dynamically simple. If ϕ has good reduction and degree at least two, then the Berkovich Julia set is the single point ζ_G , and the equilibrium measure is δ_{ζ_G} . This is very different from complex dynamics, where a degree $d \geq 2$ polynomial never has a one-point Julia set in $\mathbb{C}$.

5 Fatou and Julia sets

5.1 Classical motivation

For a rational map $R : \mathbb{P}^1(\mathbb{C}) \rightarrow \mathbb{P}^1(\mathbb{C})$, the Fatou set is the largest open set on which the iterates $\{R^n\}$ form a normal family. The Julia set is its complement. Montel's theorem is the analytic input that makes this definition powerful. Over a non-Archimedean field, one can define normality on the classical space $\mathbb{P}^1(K)$, but this misses important limiting behavior because K is totally disconnected. The Berkovich line gives the correct ambient space.

5.2 Berkovich Fatou and Julia sets

Let $\phi \in K(z)$ have degree $d \geq 2$. The chordal metric on $\mathbb{P}^1(K)$ extends to a natural path metric on suitable parts of $\mathbf{P}_{\mathrm{Berk}}^1$, and the Gel'fand topology gives a notion of equicontinuity for families of maps.

Definition 5.1. The Berkovich Fatou set $\mathcal{F}_\phi$ is the set of points $x \in \mathbf{P}_{\mathrm{Berk}}^1$ for which the family of iterates $\{\phi^n\}_{n \geq 0}$ is equicontinuous in a neighborhood of x . The Berkovich Julia set is the complement

$$\mathcal{J}_\phi = \mathbf{P}_{\mathrm{Berk}}^1 \setminus \mathcal{F}_\phi.$$

For a polynomial f , the affine Fatou and Julia sets are

$$\mathcal{F}_f^{\mathbf{A}} = \mathcal{F}_f \cap \mathbf{A}_{\mathrm{Berk}}^1, \quad \mathcal{J}_f^{\mathbf{A}} = \mathcal{J}_f \cap \mathbf{A}_{\mathrm{Berk}}^1.$$

The Fatou set is open, and the Julia set is compact and nonempty. The Julia set is backward invariant and forward invariant:

$$\phi^{-1}(\mathcal{J}_\phi) = \mathcal{J}_\phi, \quad \phi(\mathcal{J}_\phi) = \mathcal{J}_\phi.$$

These invariance properties follow from equicontinuity and the fact that ϕ is open and surjective.

Proposition 5.2. *If ϕ has potentially good reduction, then $\mathcal{J}_\phi$ consists of one type II point. Conversely, if $\mathcal{J}_\phi$ consists of one point, then ϕ has potentially good reduction.*

Proof. After conjugation, assume that ϕ has good reduction. Then the Gauss point is totally invariant. On every direction away from the Gauss point the reduction controls the dynamics, and the family of iterates is equicontinuous. Hence the only possible Julia point is ζ_G . It is indeed Julia because every neighborhood of it sees all residue directions and the reduced degree is at least two. Conversely, if the Julia set is a single point x , then total invariance of the Julia set gives $\phi^{-1}(x) = \{x\}$. Such a point must be type II, and the criterion above gives potentially good reduction. $\square$

5.3 Exceptional points

Definition 5.3. The exceptional set of ϕ is

$$E_\phi = \{a \in \mathbf{P}^1(K) : \bigcup_{n \geq 0} \phi^{-n}(a) \text{ is finite}\}.$$

Equivalently, a is exceptional if its backward orbit is finite.

The exceptional set consists only of classical points. It is small.

Proposition 5.4. *For a rational map ϕ of degree $d \geq 2$, one has*

$$\#E_\phi \leq 2.$$

If ϕ is a polynomial, then $\infty \in E_\phi$. There is a second exceptional point if and only if ϕ is conjugate to $z \mapsto z^d$.

Proof. If a is exceptional, the finite set of its backward iterates is totally invariant after replacing it by the full finite backward orbit. A rational map of degree at least two cannot have a totally invariant finite set of more than two points; this follows from the Riemann–Hurwitz formula applied to the ramification above such a set. If a polynomial is viewed as a map on $\mathbf{P}^1$, then ∞ is totally invariant, hence exceptional. If there is another exceptional point, conjugate it to 0 while keeping ∞ fixed. Then both 0 and ∞ are totally invariant, so the map has no zeros or poles away from 0 and ∞ , hence is of the form az^d . After another scaling, it is conjugate to z^d . $\square$

5.4 Fatou components

A Fatou component is a connected component of $\mathcal{F}_\phi$. Since $\mathbf{P}_{\text{Berk}}^1$ is a tree, Fatou components are connected open subtrees. A component U is periodic if $\phi^p(U) = U$ for some $p \geq 1$.

The full classification of periodic Fatou components is subtle and is one of the main topics of Benedetto’s book [1]. For this paper we only need the broad form.

Theorem 5.5 (Benedetto, qualitative form). *Let U be a periodic Fatou component for a rational map ϕ over an algebraically closed complete non-Archimedean field. Then, after replacing ϕ by an iterate, one of the following occurs.*

- (i) *U is an attracting basin: it contains an attracting periodic point, and all points of U converge to that cycle.*

(ii) U is a Rivera domain: the return map is bijective on U , and the dynamics is locally isometric in the Berkovich tree structure.

For polynomials, the basin of infinity is the most important attracting Fatou component. The Julia set is the boundary of the filled Julia set and also the support of the equilibrium measure.

6 Potential theory on $\mathbf{P}_{\text{Berk}}^1$

6.1 Measures, pushforward, and pullback

We recall the measure-theoretic language needed for equilibrium measures. Since $\mathbf{P}_{\text{Berk}}^1$ is compact Hausdorff, every finite Borel measure that appears below is Radon. A probability measure is a positive Borel measure of total mass one. The Dirac mass at a point x is denoted $[x]$ or δ_x .

Let $\phi : \mathbf{P}_{\text{Berk}}^1 \rightarrow \mathbf{P}_{\text{Berk}}^1$ be continuous. If μ is a signed Borel measure, its pushforward is

$$(\phi_*\mu)(E) = \mu(\phi^{-1}(E))$$

for Borel sets E . Equivalently,

$$\int F d(\phi_*\mu) = \int (F \circ \phi) d\mu$$

for continuous functions F .

For a rational map of degree d , the pullback of a Dirac mass is defined by

$$\phi^*[y] = \sum_{x \in \phi^{-1}(y)} \deg_x(\phi)[x],$$

and is extended linearly and continuously to finite measures. Thus

$$\int F d(\phi^*\mu) = \int \left(\sum_{x \in \phi^{-1}(y)} \deg_x(\phi) F(x) \right) d\mu(y).$$

The total mass satisfies

$$(\phi^*\mu)(\mathbf{P}_{\text{Berk}}^1) = d \mu(\mathbf{P}_{\text{Berk}}^1).$$

6.2 Gromov product

Fix a basepoint $\xi \in \mathbb{H}$. For $x, y \in \mathbf{P}_{\text{Berk}}^1$, let $x \wedge_\xi y$ be the unique point common to the three paths $[\xi, x]$, $[\xi, y]$, and $[x, y]$. This is the branch point of the tripod spanned by ξ, x, y .

Definition 6.1. The Gromov product based at ξ is

$$\langle x, y \rangle_\xi = d_{\mathbb{H}}(\xi, x \wedge_\xi y),$$

with the convention that the value is $+\infty$ if the branch point is classical.

If x is fixed, the function $y \mapsto \langle x, y \rangle_\xi$ is constant on every component of $\mathbf{P}_{\text{Berk}}^1 \setminus [\xi, x]$ and increases linearly with slope 1 along the segment from ξ to x .

6.3 Potentials and the Laplacian

Let $M(\mathbf{P}_{\text{Berk}}^1)$ be the vector space of finite signed Radon measures on $\mathbf{P}_{\text{Berk}}^1$.

Definition 6.2. For $\mu \in M(\mathbf{P}_{\text{Berk}}^1)$ and $\xi \in \mathbb{H}$, the potential of μ based at ξ is

$$u_{\mu,\xi}(x) = -\mu(\mathbf{P}_{\text{Berk}}^1) - \int_{\mathbf{P}_{\text{Berk}}^1} \langle x, y \rangle_{\xi} d\mu(y), \quad x \in \mathbb{H},$$

whenever the integral is finite. Let $\mathcal{P}$ denote the space of functions obtained in this way.

For example,

$$u_{[y],\xi}(x) = -1 - \langle x, y \rangle_{\xi}.$$

If $y = \xi$, this is the constant function -1 .

Theorem 6.3 (Baker–Rumely). *For fixed $\xi \in \mathbb{H}$, the map*

$$\mu \longmapsto u_{\mu,\xi}$$

identifies finite signed Radon measures with potentials modulo the natural integrability condition. The Laplacian is the operator

$$\Delta u_{\mu,\xi} = \mu - \mu(\mathbf{P}_{\text{Berk}}^1)[\xi].$$

Its image is the space of signed Radon measures of total mass zero, and its kernel consists of constant functions.

Proof. For atomic measures, the formula follows from the slope description of the Gromov product. The function $u_{[y],\xi}$ has outgoing slope -1 along the direction from ξ to y , and its Laplacian is $[y] - [\xi]$. Finite linear combinations give all atomic measures of total mass zero. General Radon measures are obtained by weak limits of atomic measures. The continuity of the slope formula on finite subtrees and the density of finite subtrees in the Berkovich tree extend the identity to all measures. Constants have zero Laplacian, and if $\Delta f = 0$ then all directional slopes of f on every finite subtree vanish, so f is constant. $\square$

It is often helpful to describe the Laplacian on a finite subtree $\Gamma \subset \mathbf{P}_{\text{Berk}}^1$. If f is continuous and piecewise affine on Γ , then

$$\Delta_{\Gamma} f = \sum_{p \in \Gamma} \left(\sum_{\vec{v} \in T_p \Gamma} d_{\vec{v}} f(p) \right) [p],$$

where $T_p \Gamma$ is the finite set of tangent directions of Γ at p , and $d_{\vec{v}} f(p)$ is the outgoing slope in the direction $\vec{v}$. This is the tree analogue of the second derivative.

Proposition 6.4 (Self-adjointness). *If $f, g \in \mathcal{P}$ extend continuously to $\mathbf{P}_{\text{Berk}}^1$ and their Laplacians have finite energy, then*

$$\int f d(\Delta g) = \int g d(\Delta f).$$

Proof. First suppose f and g are piecewise affine on a finite subtree Γ containing the supports of their Laplacians. By the slope formula, the left side is a finite sum over vertices. Summation by parts on each edge gives equality with the corresponding sum for $g\Delta f$; boundary terms cancel because outgoing slopes at a vertex are counted with opposite signs on adjacent edges. For general continuous potentials, approximate them uniformly on finite subtrees by piecewise affine functions. The measures have finite total variation, so the integrals converge, proving the identity. $\square$

6.4 Compatibility with dynamics

The Laplacian commutes with rational pullback and pushforward.

Proposition 6.5. *Let $\phi \in K(z)$ be nonconstant and let $f \in \mathcal{P}$. Then*

$$\phi^*(\Delta f) = \Delta(f \circ \phi), \quad \phi_*(\Delta f) = \Delta(\phi_* f),$$

where

$$(\phi_* f)(y) = \sum_{x \in \phi^{-1}(y)} \deg_x(\phi) f(x).$$

Proof. It is enough to check the identities for elementary potentials of the form $u_{[a],\xi}$, because finite linear combinations are dense and both sides are continuous under weak convergence with bounded total variation. For such a potential, the Laplacian is $[a] - [\xi]$. Pulling back gives

$$\phi^*[a] - \phi^*[\xi]$$

which is exactly the divisor of slopes of $u_{[a],\xi} \circ \phi$. The pushforward identity is similar and follows from the degree formula for fibers. $\square$

7 The equilibrium measure and equidistribution

7.1 Construction of the equilibrium measure

We now follow the potential-theoretic construction used by Favre–Rivera-Letelier [5] and presented in a systematic form in [3].

Theorem 7.1 (Equilibrium measure). *Let $\phi \in K(z)$ be a rational function of degree $d \geq 2$, and let $\xi \in \mathbb{H}$. For each $n \geq 0$, set*

$$\mu_n = d^{-n}(\phi^n)^*[\xi].$$

Then μ_n converges weakly (weak or vague in French if strictly speaking) to a probability measure μ_ϕ on $\mathbf{P}_{\text{Berk}}^1$. This measure is independent of ξ and satisfies*

$$\phi^* \mu_\phi = d\mu_\phi, \quad \phi_* \mu_\phi = \mu_\phi, \quad \mu_{\phi^m} = \mu_\phi \quad (m \geq 1).$$

Proof. Since $(\phi^n)^*[\xi]$ has total mass d^n , each μ_n is a probability measure. Put

$$\nu = d^{-1}\phi^*[\xi] - [\xi].$$

This is a signed measure of total mass zero. Hence there exists a potential $g \in \mathcal{P}$ with

$$\Delta g = \nu.$$

Using compatibility of the Laplacian with pullback,

$$\Delta(d^{-m}g \circ \phi^m) = d^{-m}(\phi^m)^*\nu = d^{-m-1}(\phi^{m+1})^*[\xi] - d^{-m}(\phi^m)^*[\xi].$$

Define

$$G_n = \sum_{m=0}^{n-1} d^{-m}g \circ \phi^m.$$

The preceding identity telescopes to

$$\Delta G_n = d^{-n}(\phi^n)^*[\xi] - [\xi] = \mu_n - [\xi].$$

Because $\xi \in \mathbb{H}$, all points in $\phi^{-1}(\xi)$ are non-classical; thus ν is supported in $\mathbb{H}$ and g extends continuously to the compact space $\mathbf{P}_{\text{Berk}}^1$. Hence g is bounded. The series

$$G_\infty = \sum_{m=0}^{\infty} d^{-m}g \circ \phi^m$$

therefore converges uniformly. Since the measures ΔG_n have total variation at most 2, the compactness theorem for potentials implies that $G_\infty \in \mathcal{P}$ and

$$\Delta G_n \longrightarrow \Delta G_\infty$$

weakly. Therefore

$$\mu_n \longrightarrow \mu_\phi := [\xi] + \Delta G_\infty.$$

The limit is a probability measure because it is a weak limit of probability measures.

Now choose another basepoint $\eta \in \mathbb{H}$ and write $[\eta] - [\xi] = \Delta h$ for a continuous potential h . Then

$$d^{-n}(\phi^n)^*[\eta] - d^{-n}(\phi^n)^*[\xi] = \Delta(d^{-n}h \circ \phi^n).$$

Since h is bounded, $d^{-n}h \circ \phi^n$ converges uniformly to 0, and the right side tends weakly to zero. Hence the limit is independent of ξ .

Finally,

$$\phi^*\mu_n = d^{-n}(\phi^{n+1})^*[\xi] = d\mu_{n+1}.$$

Passing to the limit gives $\phi^*\mu_\phi = d\mu_\phi$. Applying ϕ_* and using $\phi_*\phi^* = d$ on measures gives $\phi_*\mu_\phi = \mu_\phi$. The equality $\mu_{\phi^m} = \mu_\phi$ follows because the construction for ϕ^m uses the subsequence μ_{mn} , which has the same limit. $\square$

The measure μ_ϕ is called the equilibrium measure or canonical invariant measure of ϕ .

7.2 Polynomial specialization of the construction

For polynomials one can see the equilibrium measure more concretely through escape rates. Choose $R > 0$ so large that whenever $x = \zeta_{a,r}$ has $\text{diam}(x) > R$ or $|a| > R$, the leading term of f dominates all lower terms. Then on the branch toward infinity the map expands radii by the rule

$$\log \text{diam}(f(x)) = d \log \text{diam}(x) + O(1).$$

Consequently the sequence

$$G_n(x) = d^{-n} \log^+[T]_{f^n(x)}$$

is uniformly Cauchy on compact subtrees of $\mathbf{A}_{\text{Berk}}^1$ and converges to the escape-rate function G_f . The functional equation

$$G_f(f(x)) = dG_f(x)$$

follows directly from the definition. The filled Julia set is precisely $G_f^{-1}(0)$, while G_f is positive on the basin of infinity.

On any finite subtree Γ containing enough of the convex hull of the filled Julia set, the function G_f is piecewise affine. Its slopes record how quickly disks escape to infinity. Taking the tree Laplacian of G_f gives a signed measure of total mass zero. With the standard projective normalization this measure is

$$\Delta G_f = \mu_f - [\infty].$$

Thus μ_f may be understood as the distributional second derivative of the escape-rate function. This is exactly parallel to complex polynomial dynamics, where

$$\mu_f = \frac{1}{2\pi} \Delta G_f$$

in the plane after adding the pole at infinity. The Berkovich tree makes this analogy especially transparent: the ordinary Laplacian is replaced by the sum of outgoing slopes at vertices.

This description also clarifies why μ_f is supported on the boundary of the filled Julia set. On the interior of the filled Julia set the function G_f is identically zero, so its Laplacian vanishes there. On the basin of infinity, away from the branching locus where escape behavior changes, G_f is affine along edges, so the Laplacian again vanishes. Mass appears exactly where slopes change, namely on the Berkovich Julia set. In simple good reduction examples all the slope change is concentrated at the Gauss point; in bad reduction examples it may be distributed over a nontrivial subtree or a more complicated compact set.

7.3 Support of the equilibrium measure

Theorem 7.2. *Let $\phi \in K(z)$ have degree $d \geq 2$. Then*

$$\text{supp}(\mu_\phi) = \mathcal{J}_\phi.$$

Moreover, μ_ϕ has no atoms except in the case of potentially good reduction, where μ_ϕ is a Dirac mass at the unique type II Julia point.

Proof. We first show that $\text{supp}(\mu_\phi) \subseteq \mathcal{J}_\phi$. Let $x \in \mathcal{F}_\phi$. By equicontinuity, there is a connected affinoid neighborhood U of x on which the iterates form a normal family. One can choose a point $\xi \in \mathbb{H}$ whose full backward orbit avoids a smaller neighborhood $V \Subset U$. Indeed, if the forward orbit of the Fatou component of x does not meet some other Fatou component, choose ξ in that other component. If the component is eventually periodic, the classification of periodic Fatou components gives either an attracting basin, where one chooses ξ in an annular region not pulled back into V , or a Rivera domain, where the return map is injective and again one can avoid V .

Choose a continuous function $0 \leq F \leq 1$ which is equal to 1 on V and supported in U . Since the backward orbit of ξ avoids U ,

$$\int F d(d^{-n}(\phi^n)^*[\xi]) = 0$$

for all n . Passing to the limit gives $\int F d\mu_\phi = 0$, so $\mu_\phi(V) = 0$. Thus $x \notin \text{supp}(\mu_\phi)$.

Conversely, $\text{supp}(\mu_\phi)$ is nonempty, compact, and backward invariant because

$$\phi^* \mu_\phi = d\mu_\phi.$$

It is not contained in the exceptional set. A basic characterization of the Berkovich Julia set says that it is the smallest nonempty compact backward invariant set not contained in the exceptional set. Hence

$$\mathcal{J}_\phi \subseteq \text{supp}(\mu_\phi).$$

This proves the equality.

For atoms, suppose $\mu_\phi(\{x\}) > 0$. If the backward orbit of x is infinite, then invariance gives infinitely many disjoint sets of the same positive mass, a contradiction to total mass one. Hence x has finite grand orbit. A non-exceptional point with finite grand orbit in the Berkovich hyperbolic space forces potentially good reduction, and then the Julia set is one type II point. Conversely, in the potentially good reduction case the equilibrium measure is the Dirac mass at that point. $\square$

7.4 Equidistribution of preimages

We can now state the non-Archimedean version of Brolin's theorem, due to Favre–Rivera-Letelier [5].

Theorem 7.3 (Favre–Rivera-Letelier equidistribution). *Let $\phi \in K(z)$ have degree $d \geq 2$, and let E_ϕ be its exceptional set. If $a \in \mathbf{P}_{\text{Berk}}^1$ is not exceptional, then*

$$d^{-n}(\phi^n)^*[a] \longrightarrow \mu_\phi$$

weakly as $n \rightarrow \infty$. More generally, if ν is a probability measure on $\mathbf{P}_{\text{Berk}}^1$ with $\nu(E_\phi) = 0$, then

$$d^{-n}(\phi^n)^*\nu \longrightarrow \mu_\phi.$$

Proof sketch. The construction in Theorem 7.1 proves the result when the base point lies in $\mathbb{H}$. To pass to classical points one compares $[a]$ with a nearby non-classical point by a potential

whose singularity is controlled unless a is exceptional. If too much mass stayed away from μ_ϕ , one would obtain a nonzero limiting measure supported on the finite backward orbit of a , which is possible exactly for exceptional points. The general statement for a measure ν follows by integrating the pointwise equidistribution statement against ν and using uniform estimates outside neighborhoods of E_ϕ . $\square$

For a polynomial $f \in K[z]$ of degree $d \geq 2$, the point ∞ is exceptional. Therefore the affine version is:

$$d^{-n}(f^n)^*[a] \longrightarrow \mu_f$$

for all $a \in \mathbf{A}^1(K)$ except possibly one finite exceptional point. This is precisely the analogue of Brolin's theorem stated in the project description.

Corollary 7.4 (Affine equidistribution). *Let $f \in K[z]$ be a polynomial of degree $d \geq 2$. For $a \in K$ not exceptional, write*

$$\nu_n = \frac{1}{d^n} \sum_{f^n(b)=a} \deg_b(f^n)[b]$$

as a probability measure on $\mathbf{A}_{\text{Berk}}^1 \subset \mathbf{P}_{\text{Berk}}^1$. Then

$$\nu_n \longrightarrow \mu_f.$$

The limit is supported on the affine Julia set $\mathcal{J}_f^{\mathbf{A}}$.

Proof. The formula is exactly $\nu_n = d^{-n}(f^n)^*[a]$. Since a is not exceptional, Theorem 7.3 applies. The support statement follows from Theorem 7.2 and from the fact that the basin of infinity is contained in the Fatou set. $\square$

7.5 Characterization by invariance

Proposition 7.5. *The equilibrium measure μ_ϕ is the unique probability measure μ on $\mathbf{P}_{\text{Berk}}^1$ such that*

$$\phi^*\mu = d\mu$$

and $\mu(E_\phi) = 0$.

Proof. Let μ be such a measure. Then for every n ,

$$d^{-n}(\phi^n)^*\mu = \mu.$$

By the measure version of the equidistribution theorem, the left side converges to μ_ϕ because $\mu(E_\phi) = 0$. Hence $\mu = \mu_\phi$. $\square$

8 Entropy and ergodic theory

The paper of Favre–Rivera-Letelier [5] goes beyond equidistribution and studies ergodic properties of ϕ with respect to μ_ϕ . We summarize the main statements relevant to the present discussion.

8.1 Topological and measure entropy

Let X be a compact topological space and $T : X \rightarrow X$ continuous. The topological entropy $h_{\text{top}}(T)$ measures the exponential growth rate of distinguishable orbit segments. If μ is a T -invariant probability measure, the measure-theoretic entropy $h_\mu(T)$ measures the average information produced by the system with respect to μ .

Theorem 8.1 (Variational principle). *If X is compact and metrizable, then*

$$h_{\text{top}}(T) = \sup_{\mu} h_{\mu}(T),$$

where the supremum is over all T -invariant probability measures.

Although $\mathbf{P}_{\text{Berk}}^1$ need not be metrizable for all fields K , the entropy theory used in [5] is formulated by working with finite covers and compact invariant subtrees, which are metrizable. The tree structure gives strong entropy bounds.

Proposition 8.2. *Let $\phi \in K(z)$ have degree $d \geq 2$, and let $X \subset \mathbf{P}_{\text{Berk}}^1$ be compact, connected, and ϕ -invariant. Then*

$$h_{\text{top}}(\phi|_X) \leq \log \sup_{y \in X} \#((\phi|_X)^{-1}(y)).$$

In particular,

$$h_{\text{top}}(\phi) \leq \log d.$$

Proof. A finite connected open cover of a tree may be refined so that the number of relevant intersections is controlled by the number of boundary points. Under iteration, the number of inverse branches grows at most like the maximal number of preimages in X . Taking logarithmic growth rates of minimal subcovers gives the stated bound. $\square$

8.2 Average degree and the entropy lower bound

Define the average local degree of ϕ with respect to μ_ϕ by

$$\overline{\deg}(\phi) = \exp \left(\int_{\mathbf{P}_{\text{Berk}}^1} \log \deg_x(\phi) d\mu_\phi(x) \right).$$

Also recall the Jacobian of ϕ with respect to an invariant measure μ : it is the function $\text{Jac}_\mu \phi$ satisfying

$$\mu(\phi(E)) = \int_E \text{Jac}_\mu \phi d\mu$$

whenever ϕ is injective on E .

For the equilibrium measure, the invariance relation $\phi^* \mu_\phi = d\mu_\phi$ implies

$$\text{Jac}_{\mu_\phi} \phi(x) = \frac{d}{\deg_x(\phi)}$$

for μ_ϕ -almost every x . Rokhlin's formula gives the following result.

Theorem 8.3 (Favre–Rivera-Letelier). *Let $\phi \in K(z)$ have degree $d \geq 2$. Then*

$$h_{\mu_\phi}(\phi) \geq \log \frac{d}{\deg(\phi)}.$$

If μ_ϕ gives no mass to the Berkovich hyperbolic ramification locus, then

$$h_{\mu_\phi}(\phi) = h_{\text{top}}(\phi) = \log d_{\text{top}},$$

where d_{top} is the generic number of preimages in the Julia set.

Proof. Rokhlin's entropy formula gives

$$h_{\mu_\phi}(\phi) \geq \int \log \text{Jac}_{\mu_\phi} \phi \, d\mu_\phi.$$

Substituting the Jacobian formula yields

$$\int \log \left(\frac{d}{\deg_x(\phi)} \right) d\mu_\phi(x) = \log d - \int \log \deg_x(\phi) d\mu_\phi(x) = \log \frac{d}{\deg(\phi)}.$$

Under the stated non-ramification hypothesis the Jacobian is essentially constant on the Julia set, and the upper entropy bound is sharp. This gives equality with the topological entropy. $\square$

8.3 Zero entropy and good reduction

The following equivalences show again that potentially good reduction is the dynamically degenerate case.

Theorem 8.4 (Zero entropy criterion). *For a rational map $\phi \in K(z)$ of degree $d \geq 2$, the following conditions are equivalent.*

- (a) $h_{\text{top}}(\phi) = 0$.
- (b) $h_{\mu_\phi}(\phi) = 0$.
- (c) ϕ has potentially good reduction.
- (d) μ_ϕ is a Dirac mass at a type II point.
- (e) μ_ϕ has an atom.

Proof. The implication (a) $\Rightarrow$ (b) follows from the variational principle. If ϕ does not have potentially good reduction, then the local degree cannot equal d on the whole Julia set; after possibly passing to an iterate, the average degree is strictly smaller than d , and the lower entropy bound gives $h_{\mu_\phi}(\phi) > 0$. Thus (b) implies (c). If ϕ has potentially good reduction, conjugate so that it has good reduction. Then the Gauss point is totally invariant and the equilibrium measure is $[\zeta_G]$, giving (d). Conditions (d) and (e) are plainly related, and Theorem 7.2 shows that an atom can occur only in the potentially good reduction case. Finally a one-point Julia set has zero topological entropy, proving (d) $\Rightarrow$ (a). $\square$

9 Examples

9.1 The map $z \mapsto z^d$

Let $f(z) = z^d$ with $d \geq 2$. This polynomial has good reduction. Indeed, its homogeneous form is

$$[X : Y] \mapsto [X^d : Y^d],$$

and reduction has the same degree. Therefore the Gauss point is totally invariant:

$$f^{-1}(\zeta_G) = \{\zeta_G\}, \quad \deg_{\zeta_G}(f) = d.$$

The equilibrium measure is

$$\mu_f = [\zeta_G].$$

The Berkovich Julia set is the single point ζ_G .

On classical points the dynamics is still easy to describe. If $|z| < 1$, then $f^n(z) \rightarrow 0$. If $|z| > 1$, then $f^n(z) \rightarrow \infty$. If $|z| = 1$, then all iterates remain on the unit circle. The Berkovich point ζ_G packages the whole residue dynamics on the unit disk and is the unique source of non-equicontinuity in the Berkovich sense.

The exceptional set is $\{0, \infty\}$. For any $a \neq 0, \infty$, the d^n solutions of $z^{d^n} = a$, counted with multiplicity, equidistribute toward $[\zeta_G]$ in $\mathbf{P}_{\text{Berk}}^1$. This can seem surprising from the classical topology on K , but it is natural in the Berkovich topology: roots of all orders become dense in residue directions around the Gauss point.

9.2 Quadratic polynomials $z^2 + c$

Let $f_c(z) = z^2 + c$ over $K = \mathbb{C}_p$. If $|c| \leq 1$, then f_c has good reduction. The Gauss point is fixed, the Julia set is $\{\zeta_G\}$, and $\mu_{f_c} = [\zeta_G]$.

If $|c| > 1$, the dynamics is different. Put

$$R = |c|^{1/2}.$$

If $|z| > R$, then

$$|f_c(z)| = |z|^2 > |z|,$$

so the orbit escapes to infinity. If $|z| < R$, then

$$|f_c(z)| = |c| = R^2 > R,$$

so the next iterate escapes. Therefore any classical point with bounded orbit must satisfy $|z| = R$. On the Berkovich line, the filled Julia set is contained in the sphere of radius R , and the Julia set is a compact subtree or Cantor-type set depending on the arithmetic of c and on the residue characteristic.

This example illustrates why the Berkovich affine line is useful. The classical set of bounded points may be totally disconnected and difficult to visualize, while the Berkovich filled Julia set has a tree-theoretic convex hull on which the escape-rate function is piecewise affine. The equilibrium measure is the Laplacian of this escape-rate function, exactly paralleling the complex Green function construction.

9.3 Escape-rate functions for polynomials

For a polynomial

$$f(z) = a_d z^d + a_{d-1} z^{d-1} + \cdots + a_0, \quad d \geq 2,$$

define the escape-rate function

$$G_f(x) = \lim_{n \rightarrow \infty} d^{-n} \log^+[T]_{f^n(x)}$$

on $\mathbf{A}_{\text{Berk}}^1$. The limit exists because for sufficiently large disks the leading term dominates and the radius is essentially raised to the d -th power. The function G_f vanishes exactly on the filled Julia set and is positive on the basin of infinity. Its Laplacian, after adding the appropriate mass at infinity in $\mathbf{P}_{\text{Berk}}^1$, gives the equilibrium measure:

$$\Delta G_f = \mu_f - [\infty]$$

with the conventional normalization. This is the closest analogue of the complex formula expressing harmonic measure as the Laplacian of the Green function of the basin of infinity.

9.4 Lattes maps

Let $q \in K$ satisfy $0 < |q| < 1$. The quotient

$$E_q = K^\times / q^\mathbb{Z}$$

is a Tate elliptic curve. Multiplication by an integer m on E_q descends, after quotienting by the involution $P \mapsto -P$, to a rational map

$$L_{m,q} : \mathbf{P}_{\text{Berk}}^1 \longrightarrow \mathbf{P}_{\text{Berk}}^1$$

of degree m^2 , called a Lattes map.

The Berkovich analytification of E_q contains a skeleton isometric to a circle of length $\log |q|^{-1}$. Multiplication by m expands this skeleton by the factor m . Under the quotient to $\mathbf{P}_{\text{Berk}}^1$, the image of the skeleton becomes an interval or finite graph inside the Berkovich projective line, and the Julia set of the Lattes map is closely related to this image. The equilibrium measure is the pushforward of Haar measure on the skeleton. This example is important because the Julia set is not merely a point and the invariant measure is genuinely supported on non-classical Berkovich points.

10 Conclusion

The Berkovich affine line provides the correct topological space for non-Archimedean polynomial dynamics. Unlike the ordinary field K , it is connected, locally compact, and tree-like. Its points have a concrete classification in terms of disks and nested disks, and its hyperbolic metric turns dynamics into a form of piecewise-linear tree dynamics.

At the same time, a complete dynamical theory cannot remain purely affine. Potential theory, invariant probability measures, and entropy require compactness, and rational maps naturally

act on the projective line. Thus the correct global object is $\mathbf{P}_{\text{Berk}}^1$, with $\mathbf{A}_{\text{Berk}}^1$ sitting inside as the complement of ∞ . For polynomials, ∞ is a superattracting fixed point, and the equilibrium measure is supported on the affine filled Julia set.

The main theorem is the non-Archimedean analogue of Brolin’s equidistribution theorem. For a degree $d \geq 2$ polynomial f , the normalized preimages of any non-exceptional affine point converge to the equilibrium measure μ_f . The support of this measure is the Berkovich Julia set. When f has good or potentially good reduction, the theory degenerates in an interesting non-Archimedean way: the Julia set may be a single type II point and the equilibrium measure a Dirac mass. Otherwise, the equilibrium measure records the chaotic part of the dynamics and has positive entropy in the sense of Favre–Rivera-Letelier.

This theory opens the door to many further topics: non-Archimedean Mandelbrot sets, arithmetic equidistribution of small points, dynamics on higher-dimensional Berkovich spaces, and the interaction between Berkovich skeleta and tropical geometry. The affine line is the first case where all of these ideas can be seen explicitly.

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